

Blood Bank Management System

by

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*Project report submitted as part of the course requirements
of Database Management System (CSE 2301), Section: 05*

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SPRING 2026

EXECUTIVE SUMMARY

The Blood Bank Management System is designed to automate and organize the key operations of a blood bank through a centralized database solution. The system stores donor information, donation history, blood test records, blood inventory, hospital requests, and issue records in an integrated structure. Its purpose is to reduce manual errors, improve traceability, maintain stock availability, and ensure that only safe and tested blood units are issued.

The proposed database supports multiple users such as administrators, laboratory technicians, and hospital staff. It models real-world entities, relationships, keys, constraints, and normalization rules to ensure data consistency and integrity. The final design provides an efficient framework for donor management, blood testing, inventory control, and request processing.

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Introduction

1.1 Problem Domain Description

A blood bank is a healthcare support service that collects blood from eligible donors, performs safety screening, stores blood units under controlled conditions, and supplies compatible blood to hospitals and patients when needed. In many organizations these activities are still managed with paper files or disconnected spreadsheets. That approach often causes duplicate donor records, inaccurate stock counts, delayed response to emergency requests, weak expiry tracking, and poor traceability of issued blood units. Because blood is a life-critical resource, such data problems can directly affect patient treatment and operational efficiency.

A database-based Blood Bank Management System can solve these issues by integrating donor, donation, testing, inventory, and request information in one structured system. The database can enforce rules for donor identity, blood unit safety, stock availability, and request processing while also reducing redundancy and improving consistency.

1.2 Objectives

The objectives of the system are as follows:

1. To maintain complete and accurate donor information.
2. To record blood donation, screening, and storage details efficiently.
3. To manage blood inventory according to blood group, availability, and expiry date.
4. To process hospital blood requests in an organized and timely manner.

5. To track blood issue records for accountability and traceability.
6. To support multiple user roles with controlled access to information.

1.3 Included Processes and Excluded Processes

The proposed system includes the following processes:

1. Donor registration and profile maintenance
2. Blood donation recording
3. Laboratory testing and screening
4. Blood inventory and storage management
5. Hospital request submission and approval
6. Blood issue and distribution tracking

The proposed system excludes the following processes:

1. Online payment processing
2. Ambulance and transportation management
3. Complete patient treatment history management
4. Staff payroll and attendance management
5. Medical diagnosis and treatment recommendation

System Description & Business Rules

2.1 System/Organization Description

The Blood Bank Management System is intended for a blood bank organization that collects blood from donors, tests each donation, stores approved blood units, and supplies blood to hospitals or patients based on valid requests. The database system centralizes all operational information in one place and allows authorized users to search, update, and monitor data quickly.

The proposed system improves data accuracy, reduces manual record keeping, supports quick identification of available blood, and helps maintain regulatory safety requirements. It also improves communication between donor records, blood testing records, inventory data, and hospital requests.

2.2 Departments/Modules

The system contains the following major departments or modules:

1. **Donor Management Module:** maintains donor registration, blood group, contact information, and donation history.
2. **Laboratory Module:** records test results and determines whether blood units are safe or rejected.
3. **Inventory Management Module:** tracks available blood units, storage location, status, and expiry date.
4. **Request and Distribution Module:** manages hospital requests, issue approval, and blood issue records.

5. **Administration Module:** manages user accounts, roles, and system level reporting.

2.3 Business Rules

The business rules of the proposed system are listed below:

1. Each donor must have a unique donor ID.
2. A donor may donate blood multiple times.
3. Each donation belongs to one and only one donor.
4. One donation may produce one or more blood units or components.
5. Every blood unit must be laboratory tested before becoming available in stock.
6. A blood unit that fails screening cannot be issued.
7. Each blood unit belongs to exactly one blood group.
8. A hospital may place multiple blood requests.
9. Each blood request must be submitted by one registered hospital.
10. A blood request may contain multiple request items.
11. Only available and non-expired blood units can be issued.
12. Each test record must be created by one laboratory technician.

Data Requirements Analysis

3.1 User Roles

The main user roles of the system are Administrator, Laboratory Technician, and Hospital Staff.

3.2 Data Requirements per Role

Administrator:

1. Add, update, and delete donor records
2. Manage hospital information
3. Manage user accounts and access roles
4. View blood stock summaries by blood group
5. Monitor request and issue records
6. Generate reports for management and auditing

Laboratory Technician:

1. View donated blood unit information
2. Record screening test data
3. Update test dates and test status
4. Mark blood as safe, rejected, or pending

5. Store technician details with each test record
6. Review previous testing history

Hospital Staff:

1. Register or view hospital information
2. Submit blood requests
3. Specify required blood group and quantity
4. Track request status
5. View issue records and issue date
6. Check current blood availability

Entity–Relationship (ER) Modeling

4.1 Entities with Primary Keys

The proposed system contains the following major entities:

1. Donor (DonorID)
2. Donation (DonationID)
3. BloodUnit (UnitID)
4. Technician (TechnicianID)
5. TestRecord (TestID)
6. Hospital (HospitalID)
7. BloodRequest (RequestID)
8. RequestItem (weak entity with partial key ItemNo)
9. IssueRecord (IssueID)
10. IssueDetail (composite key IssueID, UnitID)
11. UserAccount (UserID)

4.2 Relationships with Cardinalities

The major relationships with cardinalities are as follows:

1. **Donor makes Donation (1:M):** One donor can make many donations over time, but each donation is made by only one donor.

2. **Donation produces BloodUnit (1:M):** One donation may produce one or more blood units or components, but each blood unit comes from one donation.
3. **BloodUnit has TestRecord (1:M):** One blood unit may have one or more test records if retesting is needed, but each test record belongs to one blood unit.
4. **Technician performs TestRecord (1:M):** One technician can perform many blood tests, but each test record is performed by one technician.
5. **Hospital places BloodRequest (1:M):** One hospital can place many blood requests, but each blood request is placed by one hospital.
6. **BloodRequest contains RequestItem (1:M):** One blood request can contain multiple request items, but each request item belongs to one blood request.
7. **BloodRequest results in IssueRecord (1:M):** One blood request may result in one or more issue records, but each issue record is linked to one blood request.
8. **IssueRecord includes IssueDetail (1:M):** One issue record can include many issued blood unit details, but each issue detail belongs to one issue record.
9. **BloodUnit appears in IssueDetail (1:1):** One blood unit appears in only one issue detail because a blood unit can be issued only once, and each issue detail refers to one blood unit.
10. **Administrator manages UserAccount (1:M):** One administrator can manage many user accounts, but each user account is managed under the authority of one administrator.
11. **UserAccount updates Donor records (1:M):** One user account can update many donor records, but each donor record update is performed by one user account.
12. **Inventory module tracks BloodUnit status (1:M):** The inventory module tracks the status of many blood units, while each blood unit has one current inventory status at a time.

4.3 Weak Entity with Owner

The weak entity in this system is **RequestItem**. It depends on the owner entity **BloodRequest** for its existence. The partial key of RequestItem is **ItemNo**. Therefore, the full identifying key of RequestItem is the combination of **RequestID** and **ItemNo**.

ER Diagram

5.1 Complete ER Diagram

Figure 5.1 presents the complete Entity Relationship Diagram of the Blood Bank Management System. The ER diagram includes entities, primary keys, foreign key relationships, the weak entity RequestItem, and the required cardinalities among major entities.

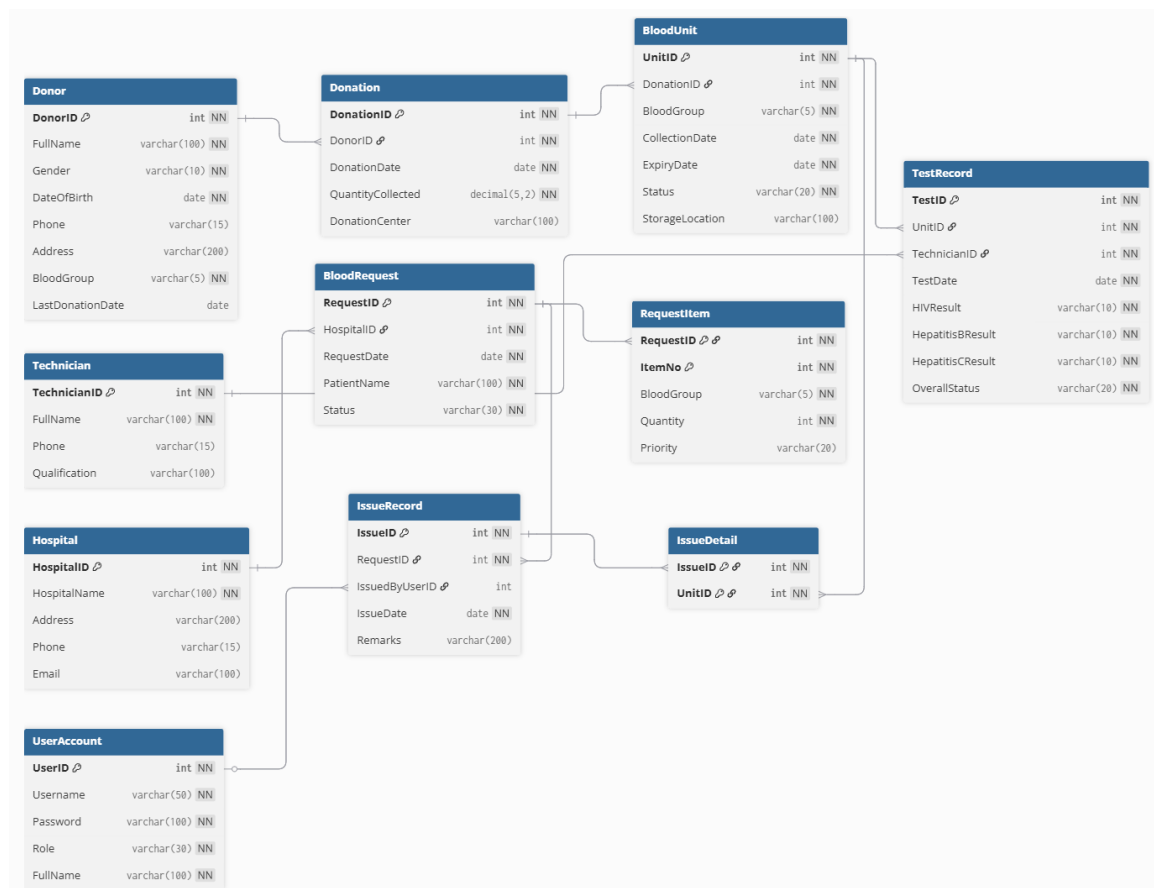


Figure 5.1: Entity Relationship Diagram of the Blood Bank Management System

Relational Schema & Keys

6.1 Tables in the Relational Schema

The relational schema of the proposed system is given below:

1. Donor(DonorID, Name, Gender, DateOfBirth, Phone, Address, BloodGroup, LastDonationDate)
2. Donation(DonationID, DonorID, DonationDate, QuantityCollected, Donation-Center)
3. BloodUnit(UnitID, DonationID, BloodGroup, CollectionDate, ExpiryDate, Status, StorageLocation)
4. Technician(TechnicianID, Name, Phone, Qualification)
5. TestRecord(TestID, UnitID, TechnicianID, TestDate, HIVResult, HepatitisBResult, HepatitisCResult, OverallStatus)
6. Hospital(HospitalID, HospitalName, Address, Phone, Email)
7. BloodRequest(RequestID, HospitalID, RequestDate, PatientName, Status)
8. RequestItem(RequestID, ItemNo, BloodGroup, Quantity, Priority)
9. IssueRecord(IssueID, RequestID, IssueDate, IssuedBy, Remarks)
10. IssueDetail(IssueID, UnitID)
11. UserAccount(UserID, Username, Password, Role, FullName)

6.2 Primary and Foreign Keys

The primary keys of the system are DonorID, DonationID, UnitID, TechnicianID, TestID, HospitalID, RequestID, IssueID, and UserID. RequestItem uses the composite key (RequestID, ItemNo), and IssueDetail uses the composite key (IssueID, UnitID).

The foreign keys are listed below:

1. Donation.DonorID references Donor.DonorID
2. BloodUnit.DonationID references Donation.DonationID
3. TestRecord.UnitID references BloodUnit.UnitID
4. TestRecord.TechnicianID references Technician.TechnicianID
5. BloodRequest.HospitalID references Hospital.HospitalID
6. RequestItem.RequestID references BloodRequest.RequestID
7. IssueRecord.RequestID references BloodRequest.RequestID
8. IssueDetail.IssueID references IssueRecord.IssueID
9. IssueDetail.UnitID references BloodUnit.UnitID

Functional Dependencies & Normalization

7.1 Functional Dependencies

The important functional dependencies for each major table are given below.

Donor

1. DonorID $\rightarrow$ Name, Gender, DateOfBirth, Phone, Address, BloodGroup, LastDonationDate
2. Phone $\rightarrow$ DonorID, Name, Address, BloodGroup (assuming phone is unique)

Donation

1. DonationID $\rightarrow$ DonorID, DonationDate, QuantityCollected, DonationCenter
2. (DonationID, DonorID) $\rightarrow$ DonationDate, QuantityCollected, DonationCenter

BloodUnit

1. UnitID $\rightarrow$ DonationID, BloodGroup, CollectionDate, ExpiryDate, Status, StorageLocation
2. UnitID $\rightarrow$ BloodGroup, ExpiryDate

Technician

1. TechnicianID $\rightarrow$ Name, Phone, Qualification
2. Phone $\rightarrow$ TechnicianID, Name, Qualification (assuming phone is unique)

TestRecord

1. TestID $\rightarrow$ UnitID, TechnicianID, TestDate, HIVResult, HepatitisBResult, HepatitisCResult, OverallStatus
2. (UnitID, TestDate) $\rightarrow$ TechnicianID, HIVResult, HepatitisBResult, HepatitisCResult, OverallStatus

Hospital

1. HospitalID $\rightarrow$ HospitalName, Address, Phone, Email
2. Email $\rightarrow$ HospitalID, HospitalName, Address, Phone (assuming email is unique)

BloodRequest

1. RequestID $\rightarrow$ HospitalID, RequestDate, PatientName, Status
2. (RequestID, HospitalID) $\rightarrow$ RequestDate, PatientName, Status

RequestItem

1. (RequestID, ItemNo) $\rightarrow$ BloodGroup, Quantity, Priority
2. (RequestID, BloodGroup) $\rightarrow$ Quantity, Priority (assuming one blood group appears once per request)

IssueRecord

1. IssueID $\rightarrow$ RequestID, IssueDate, IssuedBy, Remarks
2. (IssueID, RequestID) $\rightarrow$ IssueDate, IssuedBy, Remarks

UserAccount

1. UserID $\rightarrow$ Username, Password, Role, AdminName
2. Username $\rightarrow$ UserID, Password, Role, AdminName (assuming username is unique)

7.2 Normalization to 1NF, 2NF, and 3NF

All relations are in **First Normal Form (1NF)** because all attributes contain atomic values and there are primary key which ensure the uniqueness.

All relations are in **Second Normal Form (2NF)** because every non-key attribute depends on the primary key. In composite-key relations such as **RequestItem** and **IssueDetail**, there is no partial dependency.

All relations are in **Third Normal Form (3NF)** because there are no transitive dependencies. For example, hospital details are stored in the Hospital table instead of BloodRequest, and technician details are stored in the Technician table instead of TestRecord.

Constraints, Assumptions & Conclusion

8.1 Primary and Foreign Keys as Constraints

The database enforces all primary key and foreign key constraints mentioned in the schema. In addition, the following constraints are applied:

1. BloodGroup must be one of A+, A-, B+, B-, AB+, AB-, O+, or O-.
2. Quantity must be greater than zero.
3. ExpiryDate must be later than CollectionDate.
4. Status must be one of Available, Tested, Rejected, Issued, Expired, or Disposed.
5. Unsafe blood units cannot be marked as available.

8.2 Assumptions and Limitations

The assumptions of the proposed system are as follows:

1. Each donor has a unique donor ID.
2. Each blood unit has a unique unit ID.
3. Each blood request is submitted by a registered hospital.
4. Each blood unit can be issued only once.
5. Blood units that fail testing are never issued.

The limitations of the proposed system are as follows:

1. The system does not support online payment processing.
2. The system does not manage full patient medical history.
3. The system does not include real-time sensor-based storage monitoring.

8.3 Summary of Database Design

The proposed database design integrates donor management, blood testing, inventory control, and request processing within a structured relational model. The design includes eleven entities, well-defined relationships, primary and foreign keys, business rules, functional dependencies, and normalization up to 3NF. Therefore, the design is suitable for implementing a reliable and consistent Blood Bank Management System.

Conclusion: The Blood Bank Management System is a practical and important database application for managing donor records, blood screening, inventory, and hospital requests. The proposed design satisfies the academic requirements of the DBMS project through proper entity identification, relationship modeling, key specification, normalization, and constraint enforcement. It can serve as a strong foundation for a real-world blood bank information system.

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